

Academic Notes Template

KaoNotes

A Lightweight Template for Mathematics and Computer Science

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This chapter uses neutral example text to demonstrate the writing range of KaoNotes. Replace it with your own mathematics, computer-science notes, or technical documentation.

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1.1 Mathematics

Let $A \in \mathbb{R}^{m \times n}$, $\mathbf{x} \in \mathbb{R}^n$, and $\mathbf{b} \in \mathbb{R}^m$. A linear system can be written as

$$A\mathbf{x} = \mathbf{b}.$$

If $A = [\mathbf{a}_1 \ \cdots \ \mathbf{a}_n]$, then matrix–vector multiplication is the linear combination

$$A\mathbf{x} = x_1\mathbf{a}_1 + \cdots + x_n\mathbf{a}_n.$$

Definition 1.1.1 *The column space of A , written $\text{Col}(A)$, is the set of all linear combinations of the columns of A .*

Remark 1.1.1 The system has a solution exactly when $\mathbf{b} \in \text{Col}(A)$.

Margin notes can hold a short interpretation, reference, or calculation without interrupting the main argument.

1.2 Multilingual text

KaoNotes can keep a mainly English document while including short passages in Traditional and Simplified Chinese.

繁體中文示例：矩陣、向量與數學公式可以放在同一份學術筆記中。

简体中文示例：这一行展示简体中文字体切换。

1.3 Code beside mathematics

A short code fragment can be set like a displayed calculation:

```
import torch
A = torch.tensor([[2., 1.], [1., 3.]])
x = torch.tensor([1., 2.])
b = A @ x
```

The corresponding mathematical operation is

$$\begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 4 \\ 7 \end{bmatrix}.$$

1.4 Starting a new chapter

Create a file in the chapters directory.

Include it from `main.tex`:

```
\input{chapters/my-chapter.tex}
```

Definitions, remarks, equations, margin notes, and ordinary prose can then be combined as needed.